# Opinion Network Formation

Assignment 1: LinkedOut

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## 1 Team Name

**LinkedOut**


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## 2 GitHub Link to Code

Repository: https://github.com/dawnbliss-coder/opinion-network-formation

## 3 Dataset Documentation

### 3.1 Data Description

The dataset comprises survey responses from a class cohort regarding 60 Likert-scale statements spanning four key topics: Technology (T), Education (E), Society/Ethics (S), and Environment (V).

### 3.2 Encoding and Preprocessing

Standard deviation rescaling (without mean shift) was applied to the raw response matrix. This corrects for the difference between "extreme" and "mild" responders without destroying the genuine uniform agreement signal present across the class. Cosine similarity was used as the primary metric on the rescaled matrix as, unlike full z-scoring, which collapses neutral opinions, it preserves directional agreement effectively.

### 3.3 Handling Missing Data and Data Quality

The initial dataset contained 96 respondents. Nine respondents with more than 50% missing data (nodes 30, 39, 44, 60, 68, 73, 77, 78, 87) were dropped to ensure data quality, yielding a robust final cohort of **87 respondents**. To prevent highly skewed similarities based on only a handful of shared answers, a strict floor (`MIN_OVERLAP = 5`) was applied to pairwise comparisons: any pairwise similarity computed from fewer than 5 jointly answered statements was discarded (`np.nan`).

# 4 Pipeline Followed

## 4.1 User–User Network

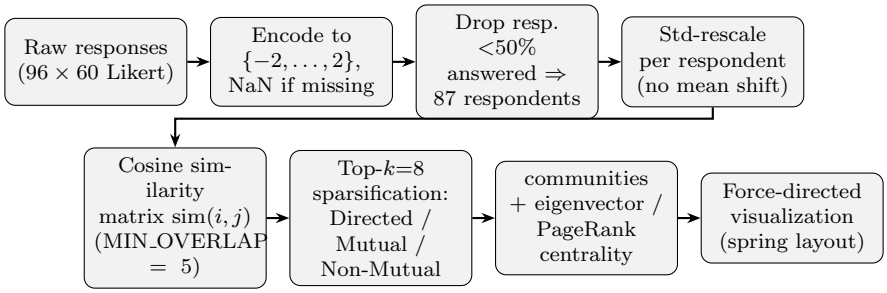



Figure 1: User–User network construction pipeline.

Because survey similarities naturally vary in strength and reciprocity, the User–User network was modeled using a deliberately designed **three-network framework**. Instead of treating one graph as universally "correct," three purpose-built views of the same underlying similarity matrix were generated, each using **top-k sparsification (k = 8)**, chosen because it produces a stable, fully connected base structure in the directed case without artificially isolating nodes.

**Directed Top-8 Network (Primary Network).** Edges are directed from user A to user B if B is in A's top 8 most similar peers. This network acts as the base representation: cosine similarity's asymmetric top-k membership is a genuine, informative feature of the data, capturing "who looks up to whom" in opinion space. It preserves the full 694 edges and is used for PageRank, in/out-strength, and connected-component metrics.

**Mutual Top-8 Network (High-Confidence Projection).** Edges exist only if A and B mutually rank each other in their top 8. This projection isolates reciprocal agreement, purposely shattering the network to reveal the deeply connected "consensus core" (the largest component) versus the fragmented periphery.

**Non-Mutual Undirected Top-8 Network (Comparison Baseline).** An edge exists if *either* A or B is in the other's top 8. This network is used specifically for cross-category and engagement-weighted comparisons. Enforcing mutuality on thinner data subsets (like specific topics) fragments the graph unevenly, so the non-mutual approach guarantees a consistently connected structure necessary for fair modularity and community comparisons.

*Methodological implementations:* Louvain community detection was used; for the primary directed graph, edge weights for reciprocal pairs were summed (weight-summed symmetrization) rather than dropped, preserving the ∼78% of network structure that is inherently one-directional. NetworkX's Eigenvector Centrality `max_iter` was raised to 1000 to ensure convergence on sparse structures. For the signed network, raw similarity is heavily right-skewed, so thresholds were dynamically calibrated to the extreme tails of the distribution: positive threshold = 0.9 (exceptional similarity) and negative threshold = −0.1 (exceptional dissimilarity).

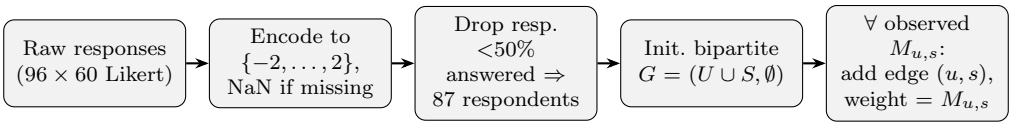



Figure 2: User–Statement network construction pipeline.

### 4.2 Statement–Statement Network

### 4.3 User–Statement Network

Formally, let $M \in \{-2, -1, 0, 1, 2, \text{NaN}\}^{U \times S}$ be the encoded, engagement-filtered response matrix ($|U|$=87 respondents, $|S|$=60 statements). The network is the bipartite graph $G = (U \cup S, E)$ with $(u, s) \in E \iff M_{u,s}$ is observed (not NaN), and edge weight $w_{u,s} = M_{u,s} \in \{-2, -1, 0, 1, 2\}$. No edge is created for a skipped/"No Comments" entry, since skipping $\neq$ a Neutral (0) response.

A reference check (`edge_threshold_table()` in `src/user_statement_network.py`) confirms that sparsifying by response magnitude does not produce a meaningfully sparser graph:

| Cutoff | Edges | Density |
|---|---|---|
| All answered (incl. Neutral) | 5,158 | 98.8% |
| Drop Neutral only | 4,491 | 86.0% |
| Strongly Agree/Disagree only | 2,349 | 45.0% |

## 5 Analysis and Visualizations

### 5.1 User–User Network

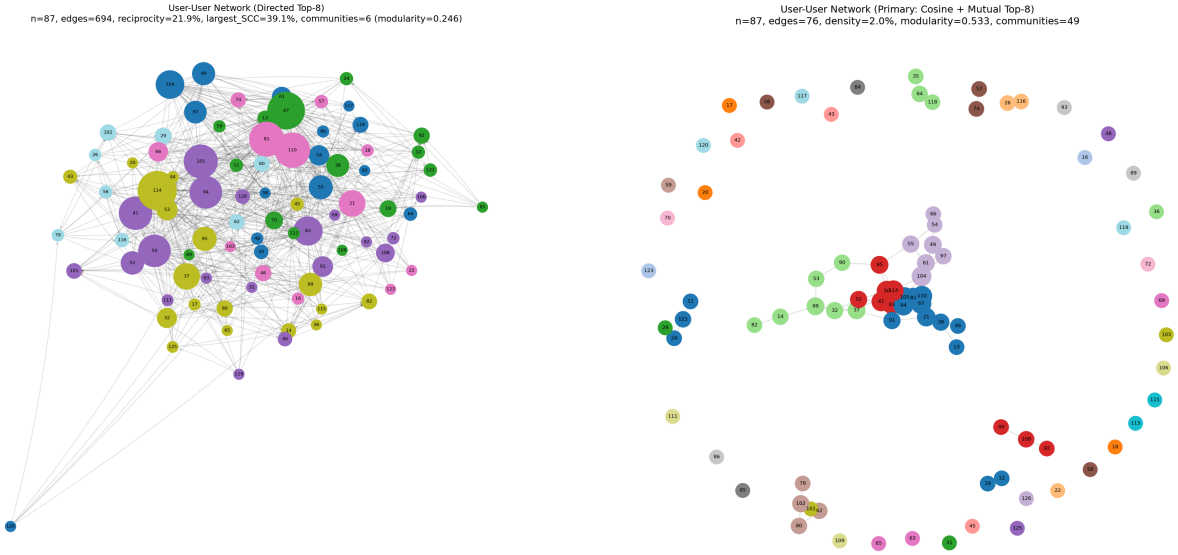



Figure 3: *Left:* Full Directed graph of top 8 similarity scores per user based on cosine similarity. *Right:* Full graph of mutual top 8 similarity scores per user based on cosine similarity.

*Topic-Specific Multiplex Layers.* To analyze how opinions cluster by topic without artificial fragmentation, the non-mutual sparsification was applied to the four distinct statement categories.

| Category | Edges | Modularity | Communities | Largest Component |
|---|---|---|---|---|
| E (Education) | 519 | 0.370 | 7 | 86 / 87 |
| S (Society/Ethics) | 621 | 0.313 | 4 | 87 / 87 |
| T (Technology) | 548 | 0.377 | 5 | 87 / 87 |
| V (Environment) | 609 | 0.299 | 7 | 86 / 87 |

*User-User Response Similarity Networks (Non-Mutual).* To ensure findings were robust regardless of the specific similarity metric, the baseline cosine similarity graph was compared against alternative network calculations.

| Metric | Edges | Modularity | Avg. Clustering | Largest CC |
|---|---|---|---|---|
| Cosine (Weighted) | 625 | 0.242 | 0.111 | 87 / 87 |
| Pearson (Unweighted) | 559 | 0.291 | 0.196 | 87 / 87 |
| Spearman (Unweighted) | 517 | 0.302 | 0.148 | 87 / 87 |
| Agreement Ratio (Raw Data) | 542 | 0.396 | 0.243 | 87 / 87 |

## 5.2 Statement–Statement Network

## 5.3 User–Statement Network

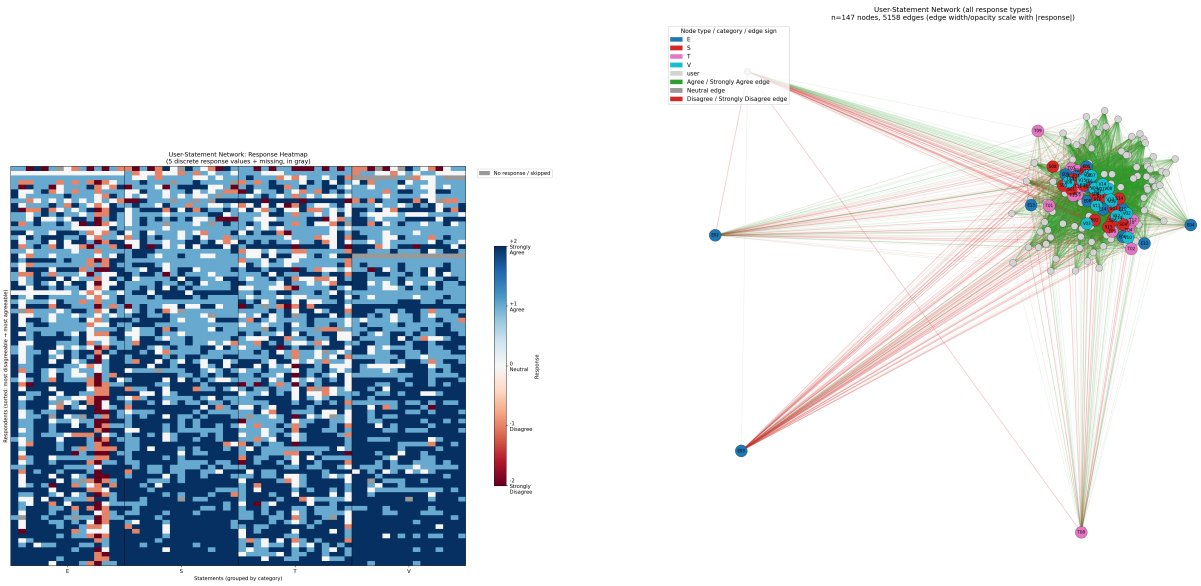



Figure 4: *Left:* response heatmap (respondents sorted most disagreeable → most agreeable; statements grouped by category; 5 response colors + gray for missing). *Right:* full bipartite network, all 5,158 edges and all five response types drawn (green = agree, red = disagree, gray = neutral; width/opacity ∝ |weight|).

| Metric | Value |
|---|---|
| Users / Statements | 87 / 60 |
| Edges (answered) | 5,158 / 5,220 possible |
| Density $= \lvert E\rvert/(\lvert U\rvert\lvert S\rvert)$ | 98.8% |
| Mean edge weight $= \frac{1}{\lvert E\rvert}\sum_{(u,s)\in E} w_{u,s}$ | +1.14 |
| Response split | 80.0% positive, 12.9% neutral, 7.1% negative |

Density near 100% means missingness does not carry much structure here. Almost everyone answered almost everything, so the real signal lives in edge *weight* rather than edge presence. The positive mean weight shows that the class leans toward agreement across the survey as a whole.

Per-node metrics: let $N(u) = \{s : (u, s) \in E\}$ (statements $u$ answered) and $N(s) = \{u : (u, s) \in E\}$ (respondents who answered $s$), so $\deg(u) = |N(u)|$, $\deg(s) = |N(s)|$.

| Metric | Formula | What it tells us |
|---|---|---|
| mean_response($u$) | $\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} w_{u,s}$ | Respondent $u$'s average stance. Used to rank who is most agreeable and who is most disagreeable overall. |
| mean_abs_response($u$) | $\frac{1}{\lvert N(u)\rvert}\sum_{s\in N(u)} \lvert w_{u,s}\rvert$ | Respondent $u$'s opinion *intensity*, independent of direction. This separates decisive respondents from wishy-washy ones (someone alternating between Strongly Agree and Strongly Disagree scores high here even if their mean_response is close to 0). |
| mean_response($s$) | $\frac{1}{\lvert N(s)\rvert}\sum_{u\in N(s)} w_{u,s}$ | Statement $s$'s consensus level and direction (near $\pm 2$ means broad agreement or disagreement, near 0 means the respondents are split). |
| std_response($s$) | $\sqrt{\frac{1}{\lvert N(s)\rvert - 1}\sum_{u\in N(s)} (w_{u,s} - \bar{w}_s)^2}$ | Spread of opinion on $s$. A low std together with a high \|mean\| points to a consensus statement, while a high std points to a polarizing or contested statement, regardless of its average. |

By these metrics, consensus statements (high mean, low std) include E15 and V13, while the most polarizing statements (high std) are E02, E03, and E04. It's worth noting that all three of the polarizing ones come from the Education category.

# 6 Results and Discussion

## 6.1 User–User Network

**A dominant consensus core.** A small, highly representative cluster of respondents acts as the ideological center of the class. Respondents 81, 105, 110, 114, and 67 consistently top the charts across both in-strength (most nominated as similar) and PageRank in the Directed network. Nodes 81, 105, 94, 67, and 110 also top eigenvector centrality in the Mutual network. They represent the "class average," acting as gravitational hubs in the opinion space.

**Hub-and-spoke agreement (low reciprocity).** Similarity in the class is fundamentally directional. Only 21.9% of directed edges are reciprocated. While the entire class is weakly connected

(100% WCC), the largest strongly connected component (SCC) is just 39% of nodes (34 nodes). Most students "point to" the consensus core, but the core does not point back. When mutuality is strictly enforced, the network shatters into 49 micro-communities with 57 unconnected nodes, showing that broad opinion camps are held together by mostly one-directional similarity.

**Polarization exists only at the extreme margins.** Using the strictly calibrated signed network (thresholds 0.9 and $-0.1$), the extreme tails of agreement and disagreement are relatively balanced but incredibly sparse: 60 positive edges vs. 47 negative edges out of thousands of possible pairs. Deep ideological opposites do exist in the class, but they are rare, disagreement overall is moderate, there are no two big heavily polarized opinion groups.

**Opinion outliers.** Looking at bottom out-strength scores in the Directed graph identifies individuals whose closest peers still aren't very similar to them. Respondent 120 (out-strength 0.23) is a massive anomaly, scoring an order of magnitude lower than the rest of the cohort (the next lowest is node 119 at 3.00). This respondent's response pattern is highly idiosyncratic compared to the broader class consensus.

**Topic-driven factionalism.** While the class shows general consensus, factionalism spikes depending on the subject matter. Topics related to Technology (modularity 0.377, 5 communities) and Education (modularity 0.370, 7 communities) display significantly higher modularity – meaning tighter, more distinct echo chambers – than Environment (0.299) or Society (0.313). This suggests that Tech and Education statements are where the most distinct dividing lines in class opinion are drawn.

### 6.2 Statement–Statement Network

### 6.3 User–Statement Network

The near-total density (98.8%) combined with a positive mean edge weight (+1.14) shows the class leans broadly toward agreement across the survey, consistent with the 80%/13%/7% positive/neutral/negative split (Section 5.3): most respondents answered most statements, and most answers were Agree or Strongly Agree.

At the respondent level, mean_response isolates a small set of genuine outliers from the agreeable majority. Respondent 120 has the lowest mean_response ($-0.21$), the only respondent whose average stance falls on the disagreement side of neutral despite answering 56 of 60 statements. That looks like a real dissenting voice rather than a low-engagement artifact – notably, this is the same respondent (120) flagged as the primary out-strength outlier in the User–User network (Section 6.1), corroborating across networks that respondent 120 is a genuine opinion outlier rather than an artifact of either method.

At the statement level, std_response identifies the Education-category statements E02, E03, and E04 as the most contested in the entire survey, while E15 and V13 are the strongest consensus statements. It's worth pointing out that **S09**, the most structurally central statement in the Statement–Statement network (highest degree and eigenvector centrality), also shows up among the most consensus statements here (mean 1.47, std 0.59). So S09's centrality seems to come from *broad shared agreement* with many other statements rather than from controversy: centrality and polarization are measuring two different things.

### 6.4 Cross-Network Synthesis

## 7 Individual Contribution

| Member | Contribution |
|---|---|
| Shreyas Agarwal | User–User network: three-network framework (Directed/Mutual/Non-Mutual Top-8 sparsification), community detection, eigenvector and PageRank centrality, signed-network threshold calibration, topic-layer and alternative-metric comparisons, corresponding report sections. |
| Name 2 | |
| Priyanka Agarwal | User–Statement network: bipartite graph construction (`user_statement_network.py`), response-weighted edge scheme, per-user/per-statement metrics, heatmap and full-network visualizations, corresponding report sections. |